Components of connective tissue

All connective tissues consist of three main components namely fibres, ground substance and cells. The **'Fibres'** of connective tissue provide support. Three types of fibres are found in the connective tissue matrix. They are **collagen**, **elastic** and **reticular** fibres. Connective tissues are of three types namely, **Loose connective tissues** (Areolar, Adipose and Reticular) and **Dense connective tissues** (dense regular, dense irregular and elastic) and **Specialized connective tissues** (cartilage, bone and blood).

Loose connective tissues

In this tissue the cells and fibres are loosely arranged in a semi fluid ground substances. For example the **Areolar connective tissue** beneath the skin acts as a support framework for epithelium and acts as a reservoir of water and salts for the surrounding body tissues, hence aptly called ***tissue fluid.*** It contains fibroblasts, macrophages, and mast cells (Figure 3.5).

Adipose tissue is similar to areolar tissue in structure and function and located beneath the skin. Adipocytes commonly called **adipose** or **fat cells** predominate and account for 90% of this tissue mass. The cells of this tissue store fats and the excess nutrients which are not utilised immediately are converted to fats and are stored in tissues. Adipose tissue is richly vascularised indicating its high metabolic activity. While fasting, these cells maintain life by producing and supplying energy as fuel. Adipose tissues are also found in subcutaneous tissue, surrounding the kidneys, eyeball, heart, etc. Adipose tissue is called '**white fat**' or **white adipose tissue.** The adipose tissue which contains abundant mitochondria is called '**Brown fat**' or **Brown adipose tissue.** White fat stores nutrients whereas brown fat is used to heat the blood stream to warm the body. Brown fat produces heat by **non-shivering thermogenesis** in neonates.

Reticular connective tissue resembles areolar connective tissue, but, the matrix is filled with fibroblasts called reticular cells. It forms an internal framework (**stroma**) that supports the blood cells (largely lymphocytes) in the lymph nodes, spleen and bone marrow.

Dense connective tissues (connective tissue proper)

Fibres and fibroblasts are compactly packed in the dense connective tissues. Orientation of fibres show a regular or irregular pattern and is called dense regular

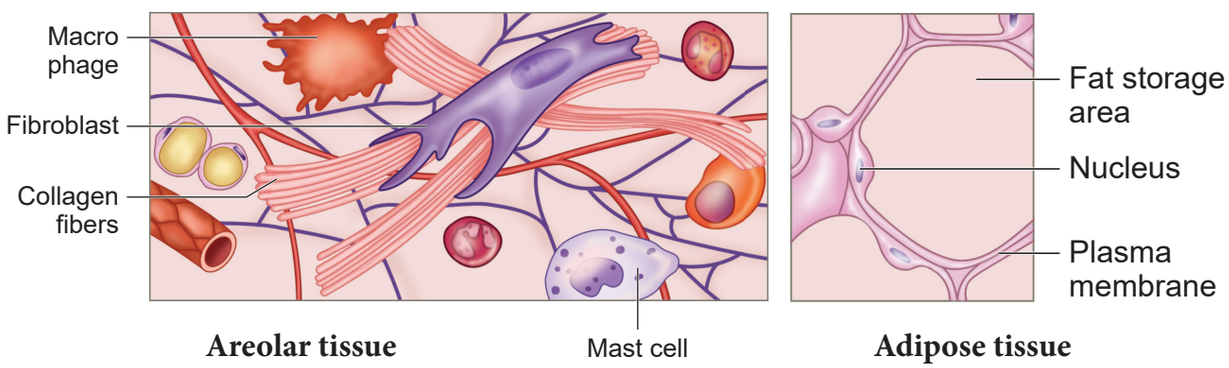


Figure 3.5 Loose connective tissues

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